

Computer Graphics

It is the use of computers to display and manipulate information in graphical or pictorial form either on visual display unit or via a printer or plotter.

→ It is an art of drawing pictures, lines, charts etc on computer screen by using programming language.

→ In computer graphics objects are presented as a collection of discrete picture element (pixel / pel)

Types – ① Non-interactive or passive

② Interactive or active.

* Open GL + Glut environment.

Computer Graphics

Pixel: Pixel is the basic building block of graphics display. We can call it picture element. Pixel is the smallest size of object or color spot that can be displayed and addressed on a screen.

Pixel $\rightarrow$ picture element.

Aspect ratio: The aspect ratio of an image is the ratio of the number of x-pixels to the number of y-pixels.

$$\text{Resolution} = 1024 \times 768$$

$$= 4:3$$

$$\text{Aspect ratio} = 1024 : 768$$

$$= 4:3$$

Image Resolution: The distance from one pixel to the next pixel is called the image resolution. The unit is called ppi (pixel per inch).

600 x 600 ppi
with round pixel or dot.

The numbers of pixels per unit length and width is referred to as resolution.

3x2 inch image resolution 300 pixel per inch.
Count total pixel?

$$\Rightarrow 3 \times 300 \times 2 \times 300$$

$$\Rightarrow 900 \times 600$$

$$\Rightarrow 540000$$

Compute the size of a 640x480 image of 240 pixel per inch.

$$= \frac{640}{240} : \frac{480}{240}$$

$$= 2.66 : 2$$

Compute the resolution of 2x2 inch image that has 512x512 pixel.

$$= \frac{512}{2} : \frac{512}{2}$$

$$= 256 : 256$$

If an image has a height of 2 inches an aspect ratio of 1:5. What is its width?

$$\Rightarrow w:h = 1:5$$

$$\Rightarrow \frac{w}{2} = 1.5$$

$$\Rightarrow w = 3$$

RGB (Additive Color Model)

R $\rightarrow$ 8 bit $\rightarrow$ 0 to 255

G $\rightarrow$ 8 bit $\rightarrow$ 0 to 255

B $\rightarrow$ 8 bit $\rightarrow$ 0 to 255

RGB image is $= 2^{24} = 16,777,216$

CMY / CMYK (Subtractive Color Model)

Yellow = R + G (1, 1, 0)

Cyan = G + B (0, 1, 1)

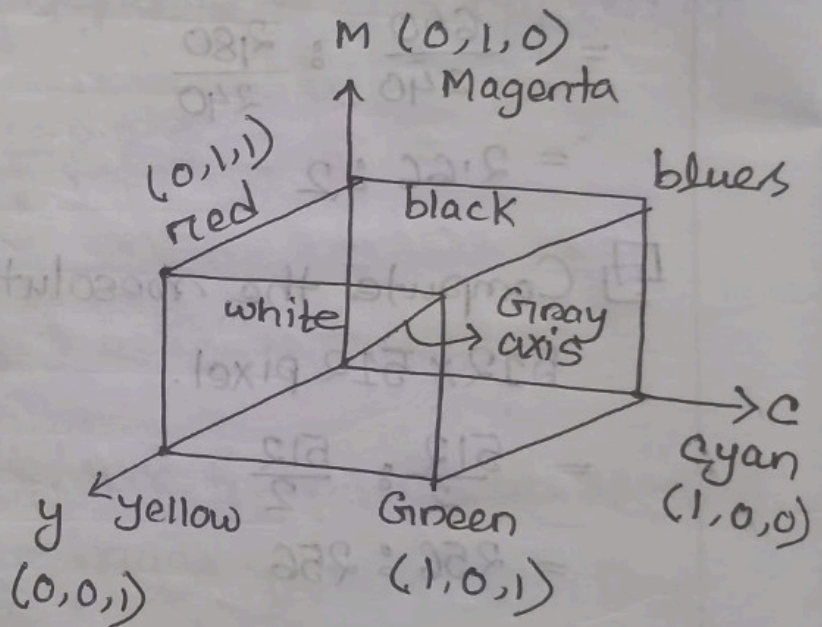
Magenta = R + B (1, 0, 1)

$$C = W - R$$

$$M = W - G$$

$$Y = W - B$$

CMY
Color space



RGB Color space

$$\begin{matrix} C \\ M \\ Y \end{matrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$

Find the CMY coordinates of color (0.2, 0.1, 0.5) in RGB.

$$\rightarrow \begin{pmatrix} C \\ M \\ Y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 0.2 \\ 0.1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.9 \\ 0.5 \end{pmatrix}$$

$x = 1 - y$

Scan conversion method: It is a process of representing graphics objects a collection of pixels.

The graphics objects are continuous. The pixels used are discrete (on or off state).

Objects:

1. Line
2. Point
3. Circle
4. arc
5. ellipse
6. rectangle
7. polygon

Computer Graphics

DDA

Given —

(x_0, y_0)

(x_n, y_n)

Step 01:

$$\Delta x = x_n - x_0$$

$$\Delta y = y_n - y_0$$

Step 02:

if $(\Delta x > \Delta y)$

steps = Δx ;

else steps = Δy ;

Step 03:

case -1
 $m < 1$

case -2
 $m = 1$

case -3
 $m > 1$

Step 04:

Repeat until endpoint.

Midpoint line drawing algorithm

Given,

Start point (x_0, y_0)

end point (x_n, y_n)

Step 01: Calculate Δx and Δy

$$\Delta x = x_n - x_0$$

$$\Delta y = y_n - y_0$$

Step 02:

$D_{initial} =$

Find the value of

$$D_{initial} = 2\Delta y - \Delta x$$

$$\Delta D = 2(\Delta y - \Delta x)$$

Step 03: The decision whether to increase x or y depends on -

Case - 1 $D_{\text{initial}} < 0$

$$x_{k+1} = x_k + 1$$

$$y_{k+1} = y_k$$

$$D_{\text{new}} = D_{\text{initial}} + 2\Delta y$$

Case - 2 $D_{\text{initial}} \geq 0$

$$x_{k+1} = x_k$$

$$y_{k+1} = y_k + 1$$

$$D_{\text{new}} = D_{\text{initial}} + 2\Delta x$$

Step 04: Repeat step 3 until endpoint.

□ Generate the points between $(20, 10)$, $(30, 18)$ using Midpoint line drawing algorithm.

Step 01: $\Delta x = 30 - 20 = 10$

$$\Delta y = 18 - 10 = 8$$

Step 02: $D_{\text{initial}} = 2 \times 8 - 10 = 6$

$$\Delta D = 2(8 - 10) = -4$$

Step 03: Case - 2 $\therefore D_{\text{initial}} > 0$,

$$x_{k+1} = 20 + 1 = 21$$

$$y_{k+1} = 10 + 1 = 11$$

$$D_{\text{new}} = 6 - 4 = 2$$

$D_{initial}$	x_{k+1}	y_{k+1}	D_{new}
—	20	10	—
6	21	11	2
2	22	12	-2
-2	23	12	14
14	24	13	6 10
6 10	25	14	6
6	26	15	2
2	27	16	-2
-2	28	16	14
14	29	17	10
10	30	18	

HW: Generate the points between (5,9), (12,16) using midpoint.

Step 01: $D_{initial} = 2 \times 8 - 10 = 6$

$\Delta D = 8(8 - 10) = -16$

Step 02: $\Delta D < 0$ Case 1: $D_{initial} > 0$

$x_{k+1} = 20 + 1 = 21$

$y_{k+1} = 10 + 1 = 11$

$D_{new} = 6 - 16 = -10$

□ Bresenham Line Drawing algorithm

Given, Start point (x_0, y_0) .

end point (x_n, y_n) .

Step 01: Calculate Δx and Δy

$$\Delta x = x_n - x_0$$

$$\Delta y = y_n - y_0$$

Step 02: Find the value of P_k

$$P_k = 2\Delta y - \Delta x$$

Step 03: The decision whether to increase x or y depends on -

Case -1

$$P_k < 0,$$

$$x_{k+1} = x_k + 1$$

$$y_{k+1} = y_k$$

$$P_{k+1} = P_k + 2\Delta y$$

Case -2

$$P_k \geq 0,$$

$$x_{k+1} = x_k + 1$$

$$y_{k+1} = y_k + 1$$

$$P_{k+1} = P_k + (2\Delta y - 2\Delta x)$$

* Difference between
Bresenham and DDA.

Serial No.	Variable Name	Type
01	total	float
02	Capital	float
03	rate	float

Assignment

Peripheral CRT

LCD

LED

Random scan display

Raster scan display

Lookup table.

Circle drawing Algorithm (Bresenham):

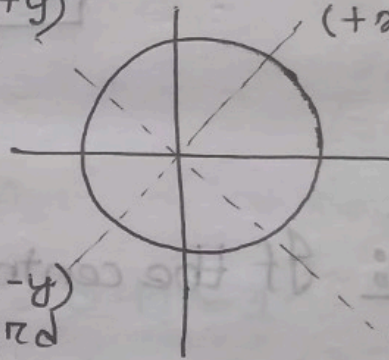
8 way symmetric

$(-x, +y)$
2nd

$(+x, +y)$
1st

$(-x, -y)$
3rd

$(+x, -y)$
4th



Given,

Centre point of circle (x_0, y_0)

Radius of circle $= R$

Step 01: Assigning starting point coordinate,

$$x_0 = 0$$

$$y_0 = R$$

* What is 8way symmetry of circle?

Step 02: Calculate decision parameter P_k

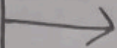
$$P_k = 3 - 2R$$

Step 03: Current point x_k, y_k and next point x_{k+1} and y_{k+1} depends on P_k .

Two case:

Case: 1

$$P_k < 0$$



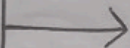
$$x_{k+1} = x_k + 1$$

$$y_{k+1} = y_k$$

$$P_{k+1} = P_k + 4(x_{k+1} + y_{k+1}) + 6$$

Case: 2

$$P_k \geq 0$$



$$x_{k+1} = x_k + 1$$

$$y_{k+1} = y_k - 1$$

$$P_{k+1} = P_k + 4(x_{k+1} - y_{k+1}) + 10$$

✍

Step 04: If the centre point (x_0, y_0) is not $(0, 0)$

then —

$$x_{\text{plot}} = x_c + x_0$$

$$y_{\text{plot}} = y_c + y_0$$

[x_c and y_c denotes the current value of x on Yaxis]

Step 05: Keep repeating 3 and 4 until

$$x_{\text{plot}} \geq y_{\text{plot}}$$

Step 06: Generate all the points of one octate and find the other

Given, the centre point of a circle $(0,0)$ and radius 8.
Generate all the points of circle.

Solution: Given, $(x_0, y_0) = (0,0)$

$$R = 8$$

Step 01:

$$x_0 = 0$$

$$y_0 = 8$$

Step 02:

$$P_0 = 3 - 2R$$

$$= 3 - (2 \times 8) = -13$$

Step 03:

$$\therefore P_0 < 0$$

Case 1:

$$x_{k+1} = x_k + 1 = 0 + 1 = 1$$

$$y_{k+1} = y_k = 8$$

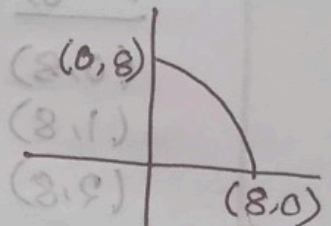
$$P_{k+1} = P_k + 4(x_{k+1}) + 6 = -3$$

Step 04:

Not Applicable

Step 05:

P_k	P_{k+1}	(x_{k+1}, y_{k+1})
		$(0, 8)$
-13	-3	$(1, 8)$
-3	11	$(2, 8)$
11	5	$(3, 7)$
5	7	$(4, 6)$
7		$(5, 5)$
		$(6, 4)$
		$(7, 3)$



		(8, 2)
		(8, 1)
		(8, 0)

Quadrant 1

(x, y)

(0, 8)

(1, 8)

(2, 8)

(3, 7)

(4, 6)

(5, 5)

(6, 4)

(7, 3)

(8, 2)

(8, 1)

(8, 0)

Quadrant 4

(x, -y)

(0, -8)

(1, -8)

(2, -8)

(3, -7)

(4, -6)

(5, -5)

(6, -4)

(7, -3)

(8, -2)

(8, -1)

(8, 0)

Quadrant 2

(-x, +y)

(0, 8)

(-1, 8)

(-2, 8)

(-3, 7)

(-4, 6)

(-5, 5)

(-6, 4)

(-7, 3)

(-8, 2)

(-8, 1)

(-8, 0)

Quadrant 3

(-x, -y)

(0, -8)

(-1, -8)

(-2, -8)

(-3, -7)

(-4, -6)

(-5, -5)

(-6, -4)

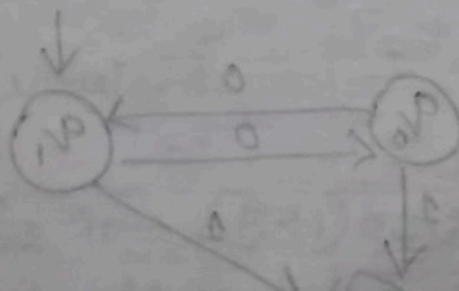
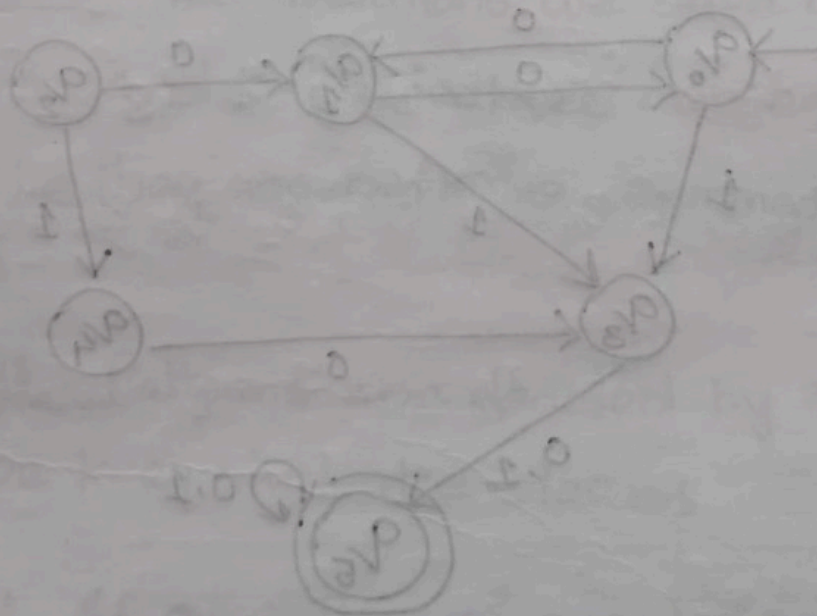
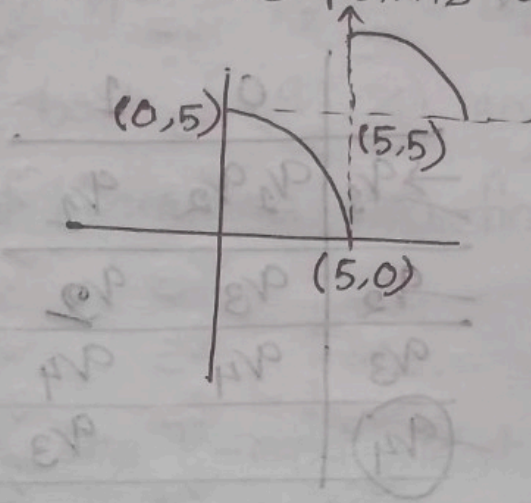
(-7, -3)

(-8, -2)

(-8, -1)

(-8, 0)

Given, the center point of a circle $(5,5)$ and radius 5. Generate all the points of circle.



2D - Transformation

Changing an object after creating in terms of position or size

Change position / shape

① Translation.

② Rotation.

③ Reflection.

④ Scaling.

⑤ Shearing

} position change of object

} size

① Translation

It will shift the object from the position to other position

→ To translate a point from coordinate position (x, y) to another (x_1, y_1) are added algebraically then the translation distance:

$$x_1 = x + T_x$$

$$y_1 = y + T_y$$

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} T_x \\ T_y \end{bmatrix}$$

$$P' = P + T$$

$(T_x \text{ \& } T_y)$ - shift vectors

$a(2,2)$, $b(10,12)$, $c(5,5) \rightarrow$ Translate the triangle with $T_x = 5$ and $T_y = 6$.

$$a' = \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} T_x \\ T_y \end{bmatrix} \\ = \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$b' = \begin{bmatrix} 15 \\ 18 \end{bmatrix} \quad c' = \begin{bmatrix} 10 \\ 11 \end{bmatrix}$$

(11) Rotation

It is a process of changing the angle of the object.

— Rotation can be clockwise or anti-clockwise.

Let, coordinate p in polar form.

$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$x' = r \cos(\phi + \theta) \\ = r (\cos \phi \cos \theta - \sin \phi \sin \theta) \\ = r \cos \phi \cos \theta - r \sin \phi \sin \theta \\ = x \cos \theta - y \sin \theta.$$

$$y' = r \sin(\phi + \theta) \\ = r (\sin \phi \cos \theta + \cos \phi \sin \theta) \\ = r \sin \phi \cos \theta + r \cos \phi \sin \theta \\ = x \sin \theta + y \cos \theta.$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Q On triangle $A(2, 2)$, $B(8, 2)$, $C(5, 5)$, Rotate to 60°
 90° .

$$\begin{bmatrix} 7 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 10 \\ 11 \end{bmatrix} = \begin{bmatrix} 2 \\ 18 \end{bmatrix}$$

① Rotation

It is a process of changing the angle of the object.
 — Rotation can be clockwise or anti-clockwise.
 Let coordinates P in polar form.

$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$x = r \cos(\phi + \theta)$$

$$r \cos(\phi + \theta) = x$$

$$r \cos \phi \cos \theta - r \sin \phi \sin \theta = x$$

$$r \cos \theta - y \sin \theta = x$$

$$y = r \sin(\phi + \theta)$$

$$r (\sin \phi \cos \theta + \cos \phi \sin \theta) = y$$

$$r \sin \phi \cos \theta + r \cos \phi \sin \theta = y$$

$$x \sin \theta + y \cos \theta = r \sin \theta$$

Reflection

Reflection is a kind of rotation where the angle of rotation is 180° .

The reflected object is always formed on the other side of "Mirror".

The size will be same.

Consider, a point object O has to be reflected in 2D plane.

Let, initial coordinate of O = (x_{old}, y_{old})

New coordinate will be —

(x_{new}, y_{new})

Reflection on X axis:

The reflection is achieved by using —

$$x_{new} = x_{old} \quad , \quad y_{new} = -y_{old}$$

The matrix form will be —

$$\begin{bmatrix} x_{new} \\ y_{new} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} x_{old} \\ y_{old} \end{bmatrix}$$

For y axis,

$$x_{new} = -x_{old}$$

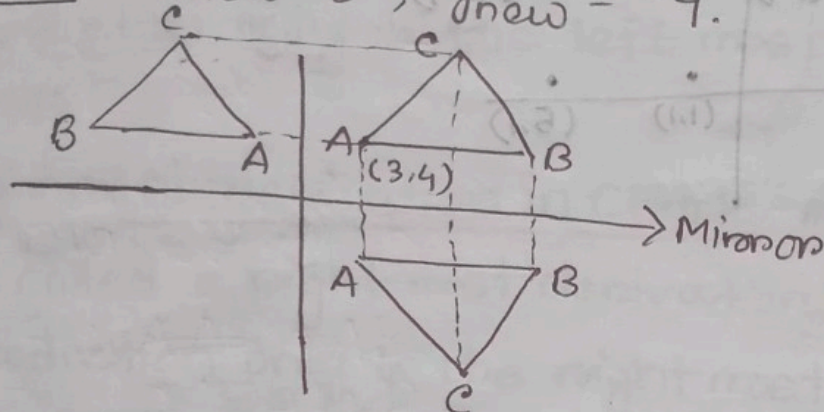
$$y_{new} = y_{old}$$

The matrix form will be —

$$\begin{bmatrix} x_{new} \\ y_{new} \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} x_{old} \\ y_{old} \end{bmatrix}$$

Given, triangle $A(3,4)$, $B(6,4)$, $C(5,6)$. Apply reflection on x -axis.

For A: $x_{\text{new}} = 3$, $y_{\text{new}} = -4$.



Scaling:

It is used to increase or decrease the size of object.

Scaling factor determines whether the object size is to be increased or reduced.

If S_x or $S_y > 1$, then increase.

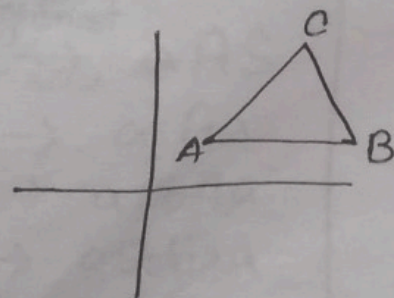
If S_x or $S_y < 1$, then reduce.

$(x_{\text{old}}, y_{\text{old}})$

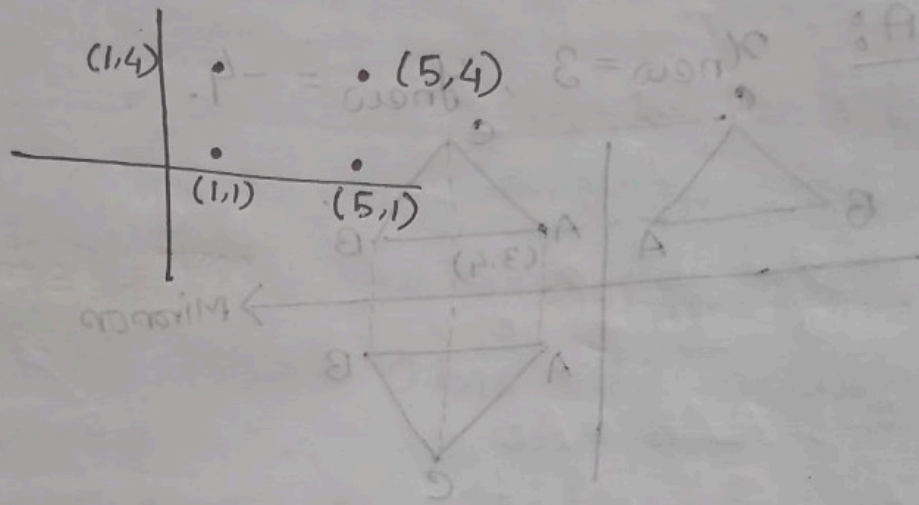
$$x_{\text{new}} = x_{\text{old}} \times S_x$$

$$y_{\text{new}} = y_{\text{old}} \times S_y$$

$$\begin{bmatrix} x_{\text{new}} \\ y_{\text{new}} \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} x_{\text{old}} \\ y_{\text{old}} \end{bmatrix}$$



Given, square A(1,1), B(1,2), ^{5,1}(~~4,5~~), ^{5,4}(~~4,4~~).
 Scaling factor towards $x=3$, for $y=3$



2D

Shearing:

X-axis:

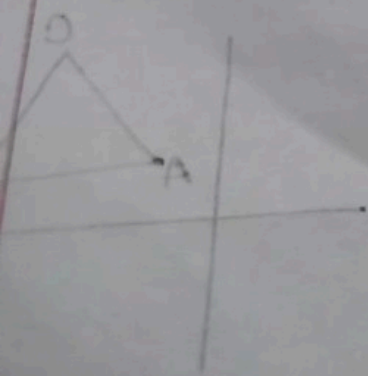
$$X_{\text{new}} = X_{\text{old}} + Sh_x \times Y_{\text{old}}$$

$$Y_{\text{new}} = Y_{\text{old}}$$

Y-axis:

$$X_{\text{new}} = X_{\text{old}} + Sh_y \times Y_{\text{old}}$$

$$Y_{\text{new}} = Y_{\text{old}}$$

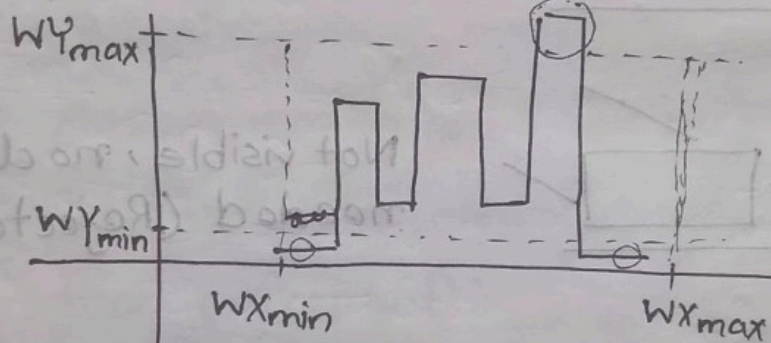


$$\begin{bmatrix} X_{\text{old}} \\ Y_{\text{old}} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} X_{\text{new}} \\ Y_{\text{new}} \end{bmatrix}$$

Clipping: Cutting the unnecessary part. One window is given the coordinate (WX_{min}, WY_{min}) & (WX_{max}, WY_{max}) is called clipping window.

Clipping application:

- It will extract the part we desire.
- To identify visible and invisible part in 3D object.
- For drawing operation.
- For deleting, copying, moving part of an object.

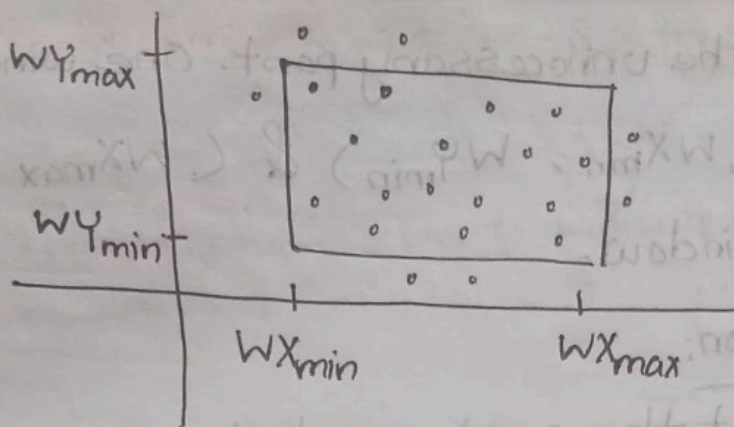


Types:

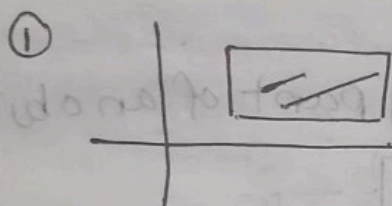
- Point clipping
- Line clipping
- Polygon clipping
- Curve clipping
- Text clipping

① Point clipping: Satisfy 4 condition otherwise can't see the points.

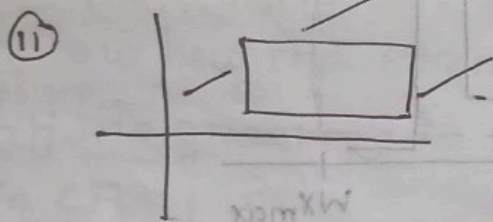
$$\begin{aligned} X &\geq WX_{min} & Y &\geq WY_{min} \\ X &\leq WX_{max} & Y &\leq WY_{max} \end{aligned}$$



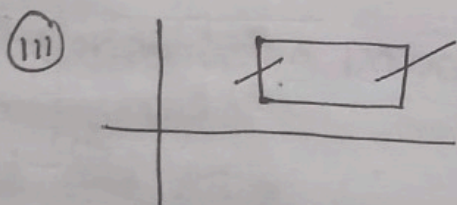
② Line Clipping:



fully visible, no clipping needed (Accepted)



Not visible, no clipping needed (Rejected).



Partial visible, clipping needed, unnecessary part will be cut off. (clipping).

☐ Cohen Sutherland Line Clipping Algorithm:

All lines come under any of these case -

① Accepted ② Rejected ③ Clipping.

③ Clipping Case: In this case, the line should be clipped. The category of the line is

found based on nine regions. All nine regions are assigned with codes. Each code is 4 bits. If both endpoint ~~bits~~ bits of the line is zero, then it is visible.

1	2	3
1001	1000	1010
4	5	6
0001	0000	0010
7	8	9
0101	0100	0110

TBRL-method

Top
Bottom
Right
Left

Clipping Algorithm:

Step-1: Assign region code for each end points.

Step-2: If both end point has region code (0000) then (accepted).

Step-3: Else perform logical AND operation for both region code.

3.1: If the result is not (0000), then rejected.

3.2: ~~Choose~~ Else need clipping.

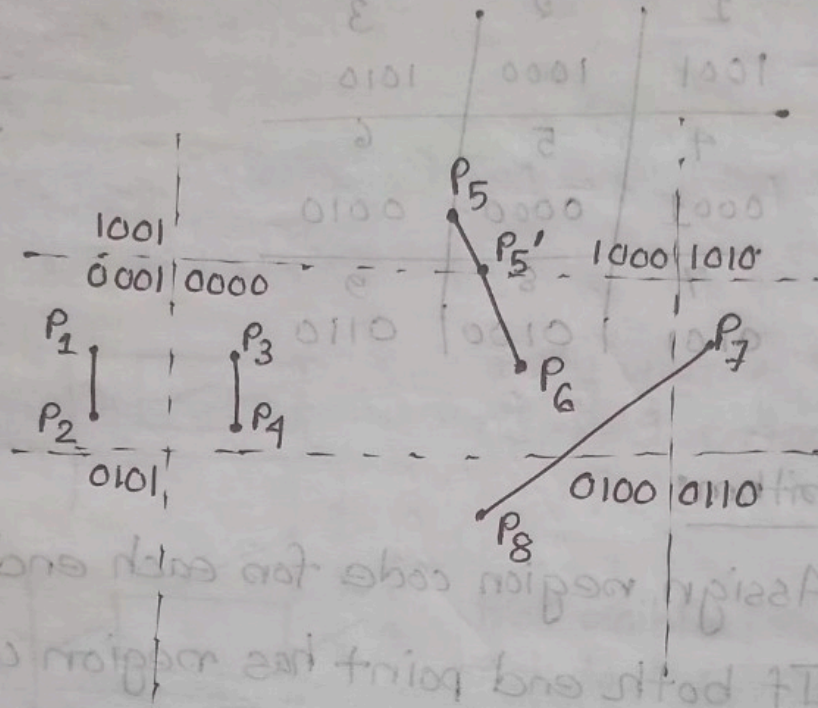
3.2.1: Choose an endpoint that is outside the window.

3.2.2: Find the intersection point at the window boundary based on region code.

3.2.3: Replace the intersection point with endpoint and update code.

3.2.4: Repeat step-2 until find clipped
Accepted line.

Step-4: Repeat Step 1 for other line.



Case-1 $P_1 \& P_2$

P_1 0 0 0 1 (non zero)

P_2 0 0 0 1 (non zero)

AND 0 0 0 1 (non zero)

$P_1 - P_2$ (Rejected)

Case - 2 $P_3 \& P_4$

P_3 0 0 0 0 (zero)

P_4 0 0 0 0 (zero)

AND 0 0 0 0 (zero)

$P_3 - P_4$ (Accepted)

Case-3 P_5 & P_6

P_5 1000 (non zero)

P_6 0000 (zero)

AND 0000 (zero)

(Clipping)

P_5' 0000

P_6 0000

0000

$P_5 - P_5'$

$P_5' - P_6$ (Accepted)

Case-4 P_7 & P_8

P_7 0010 (non zero)

P_8 0100 (non zero)

AND 0000 (zero)
(clipping)

Clip $P_7 - P_7'$

$P_7' - 0000$ (zero)

$P_8 - 0100$ (non zero)

AND 0000 (zero)
(clipping)

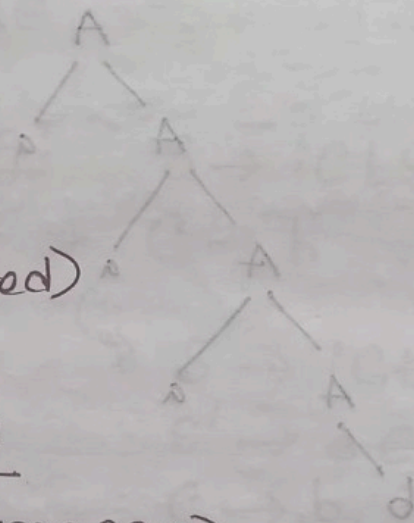
Clip $P_8 - P_8'$

P_7' 0000

P_8' 0000

AND 0000

(Accepted)



- * Cohen-Sutherland Polygon Clipping Algo
- * Weiler-Athenston Polygon Clipping Algo
- * Text Clipping

Computer Graphics - (5/5/26)

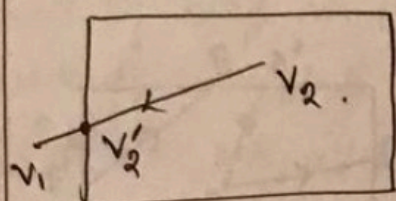
Date: / /

Cohen Sutherland Polygon Clipping Algorithm

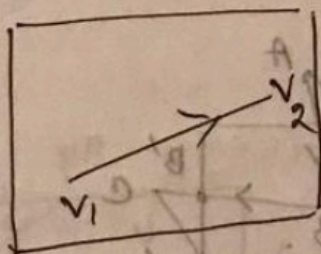
condition 4 case -

1. left clip
2. Right
3. Top
4. Bottom.

in-out



in - in



Output

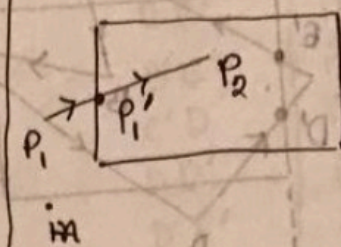
$V_2' V_1$
the direction
of the line was
from V_2 to V_1

Output

$V_1 V_2$

condition 4.

Out-in

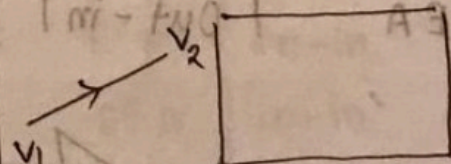


Output

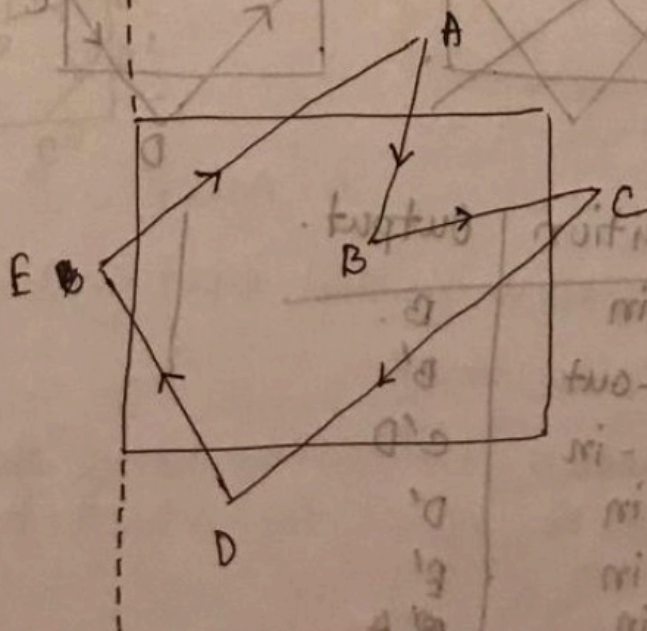
$P_1' P_2$

↳ the direction
of the line was
from P_1 to P_2

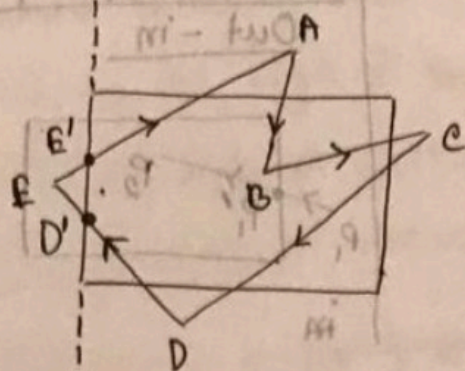
Out-Out



Output
Nothing

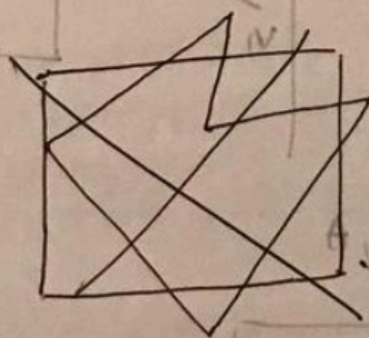


Output



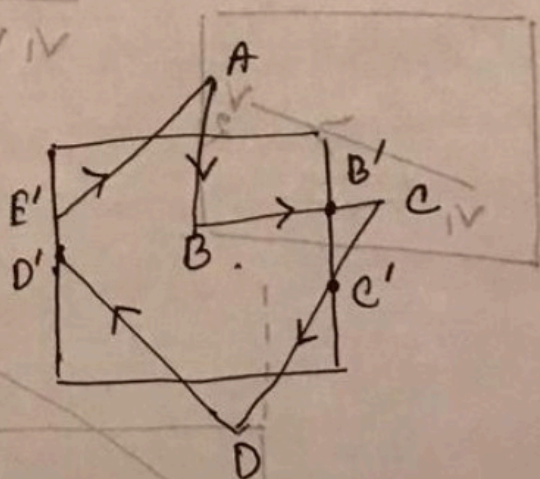
Left Clip

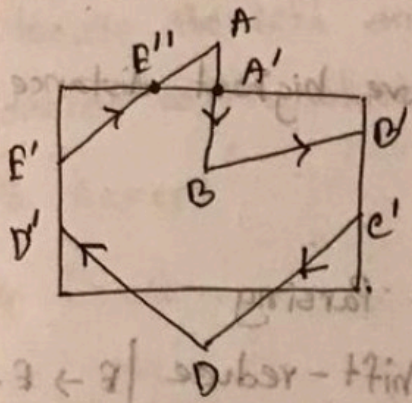
Vertex	Condition	Output
AB	in-in	B
BC	in-in	C
CD	in-in	D
DE	in-out	D'
EA	Out-in	A



Right Clip

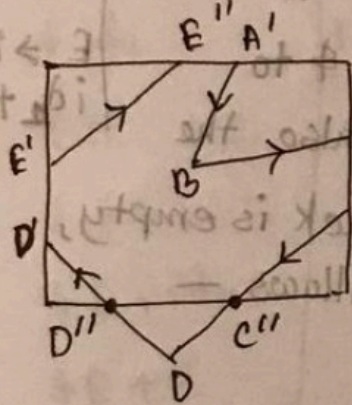
Vertex	Condition	Output
AB	in-in	B
BC	in-out	B'
CD	out-in	C'D
DD'	in-in	D'
D'E'	in-in	E'
E'A	in-in	A





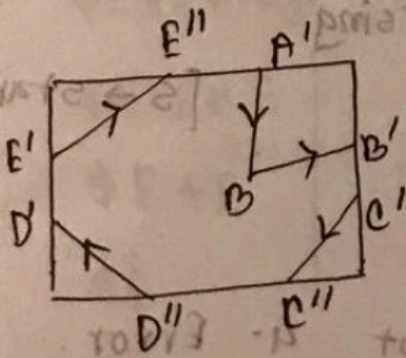
Top Clip

Vertex	condition	Output
AB	Out-in	A'B
BB'	in-in	B'
B'C'	in-in	C'
C'D	in-in	D'
DD'	in-in	D'
D'E'	in-in	E'
E'A	in-out	E''



Bottom Clip

Vertex	condition	Output
A'B	in-in	A'B
BB'	in-in	B'
B'C'	in-in	C'
C'D	in-out	C''
DD'	out-in	D''D'
D'E'	in-in	E'
E'E''	in-in	E''
E''A	in-in	A'

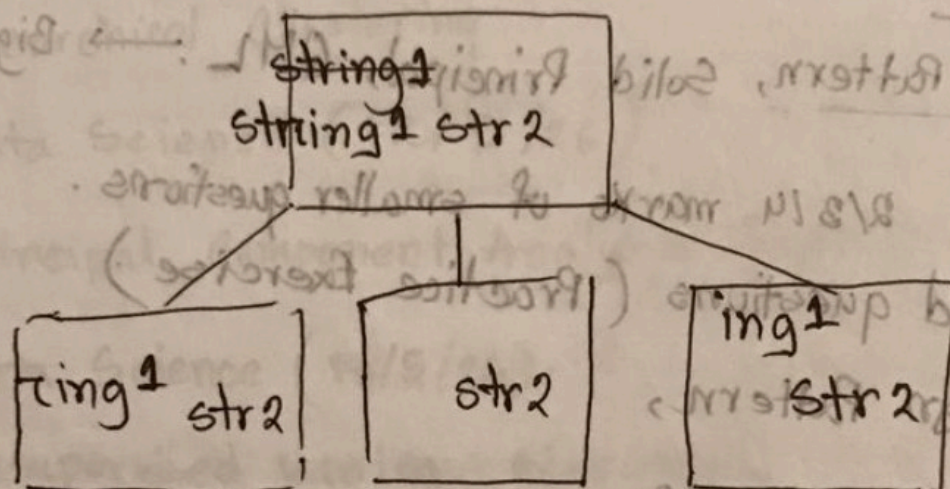
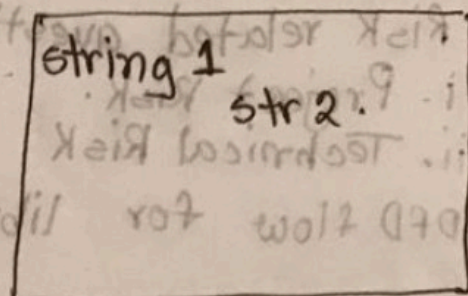


Computer Graphics (20/5/26)

Atherton Polygon Clipping Algorithm ** Exam

Text Clipping —

- All or none string clipping
- All or none character
- Text



Computer Graphics (15/06/26)

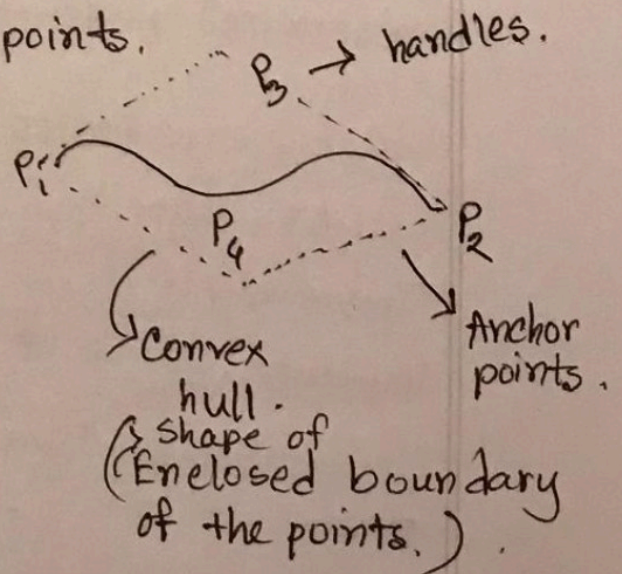
Curve -

A curve is an infinitely large set of points. Here, each point has ~~the~~ two ~~ex~~ neighbours except ~~and~~ end points.

Curve

- ↳ Implicit Curve
- ↳ Explicit curve
- ↳ Parametric Curve

definition.



Bell Curve —

Bezier Curve —

The curves can be generated under control points. Bezier curve is a parametric ~~cur~~ curve defined by a set of control points.

Simple Bezier Curve
Control = 2
degree = 2 - 1

Quadratic
Control = 3
degree = 3 - 1 = 2

Cubic
control = 4
degree = 4 - 1 = 3

Properties —

1. They always pass through the first and last control points.
2. They are contained in the convex hull of the defining control points.
3. The degree of the polynomial defining curve segments is one less than the control points.
4. Here the control points can alter the shape.

Equation —

$$P(t) = \sum_{i=0}^n B_i J_{n,i}(t)$$

Here, t is any parameter where $0 \leq t \leq 1$.

$P(t)$ = Any point lying on Bezier curve.

n = degree

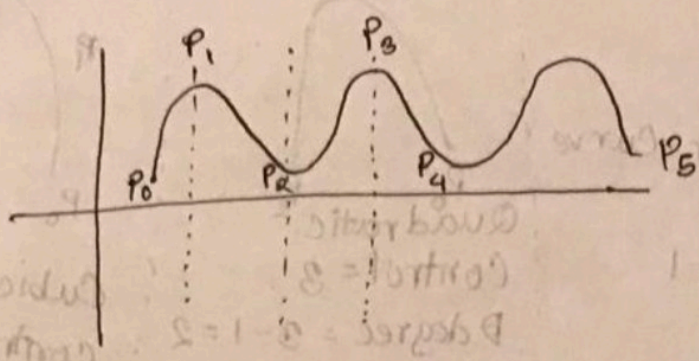
B_i = i th control points

$J_{n,i}(t)$ = Binding function:

$$= c(n, i) = t^i (1-t)^{n-i}$$

where, $c(n, i) = \frac{n!}{i!(n-i)!}$

B-spline curve



$K=3$

Segment

Control Points

Q_1

$P_0 P_1 P_2$

Q_2

$P_1 P_2 P_3$

Q_3

$P_2 P_3 P_4$

Q_4

$P_3 P_4 P_5$

Topics -

1. Hidden Surface
2. Projection
3. Parallel " { Difference
4. Perspective " { definition, theory.
5. 3D transformation. 1, 2 marks definition, theory
6. Painter's Algorithm
- 7.